Calculus Prac1

Zadanie	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Punkty															
Zadanie	**15**	**16**	**17**	**18**	**19**	**20**									**Suma**
Punkty															

Zadanie 1:

25. Water is flowing out of a reservoir at $300t^2$ liters per second for t between 0 and 5. How many liters are released in this period?

Zadanie 2:

47. One positive number plus the square of another equals 48. Choose the numbers so that their product is as large as possible.

Zadanie 3:

84. Verify the horserace theorem for $f_1(x) = x^2 + x - 2$ and $f_2(x) = x^3 + 3x^2 - 2x - 2$ on $[0, 1]$.

Zadanie 4:

8. Where is $(x^4 + 1)/(x^3 - 8)$ continuous?

Zadanie 5:

34. Suppose that f is a continuous function on $[a, b]$ and that $f(x) \neq 0$ for all x in $[a, b]$. Assume that $a \neq b$ and that $f((a + b)/2) = 1$. Prove that $\int_a^b f(x)\,dx > 0$. [*Hint*: Find a lower sum.]

Zadanie 6:

58. Show that the power of a function rule for negative powers follows from that rule for positive powers and the reciprocal rule.

Zadanie 7:

21. A sphere of radius r has volume $V = \frac{4}{3}\pi r^3$. What is the rate of change of the volume of the sphere with respect to its radius? Give a geometric interpretation of the answer.

Zadanie 8:

33. Read Example 8. Show that for any conical tank, the ratio of dh/dt to the rate of rainfall is equal to the ratio between the area of the tank's opening to the area of the water surface. (In fact, this result is true for a tank of any shape; see Review Exercise 32, Chapter 9.)

Zadanie 9:

In Exercises 1–8, assume that x and y are functions of t. Relate dx/dt and dy/dt using the given relation.

1. $x^2 - y^2 = 3$

2. $x + y = 4$

3. $x - y^2 - y^3 = 4$

4. $8x^2 + 9y^2 = 5$

5. $x + y^2 = y$

6. $\sqrt{x + 2y} = x$

7. $\sqrt{x} - \sqrt{y} = 5$

8. $(x^2 + y^2 + y^3)^{3/2} = 5$

Zadanie 10:

51. A factory employing w workers produces $100w + w^2/100 - (1/5000)w^4$ dollars worth of tools per day. Find the marginal productivity of labor when $w = 20$.

Zadanie 11:

Find linear approximations for the expressions in Exercises 49–52.

49. $\sqrt[4]{15.97}$

50. $(4.02)^{3/2}$

51. $(-26.98)^{2/3}$

52. $\sqrt[2]{122}$

Zadanie 12:

15. During takeoff, a 747 has $25{,}000 - 80t + 2t^2 + 0.2t^3$ gallons of fuel in its tanks t seconds after starting its takeoff, $0 \leqslant t \leqslant 10$. How many gallons per second are being burned 2 seconds into the takeoff?

Zadanie 13:

26. Tell where the function in Fig. 3.2.22 is increasing and where it is decreasing.

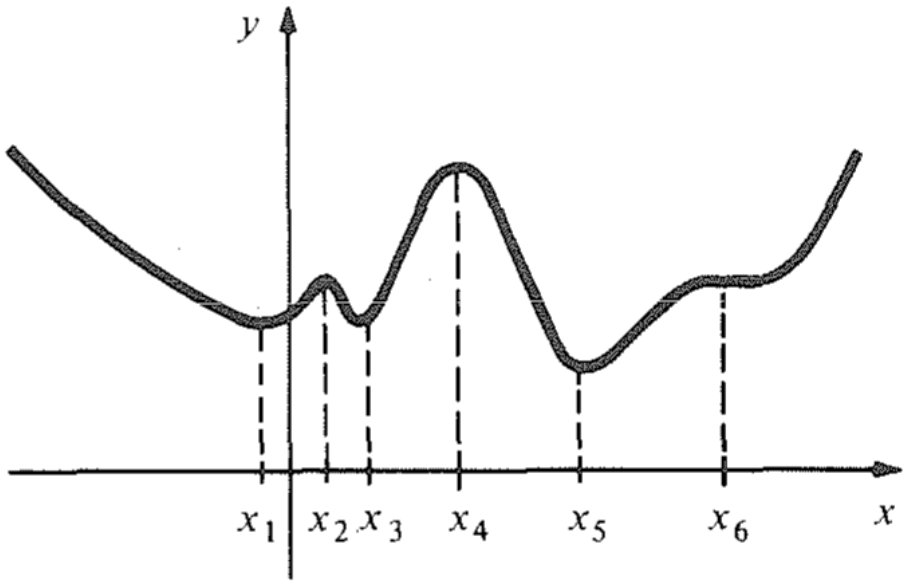


Figure 3.2.22. Find the local maxima and minima.

Zadanie 14:

37.

(a) At time $t = 0$, a container has 1 liter of water in it. Water is poured in at the rate of $3t^2 - 2t + 3$ liters per minute (t = time in minutes). If the container has a leak which can drain 2 liters per minute, how much water is in the container at the end of 3 minutes?

(b) What if the leak is 4 liters per minute?

(c) What if the leak is 8 liters per minute? [*Hint*: What happens if the tank is empty for a while?]

Zadanie 15:

19. The curves $y = x^3$ and $y = x$ divide the plane into six regions, only two of which are bounded. Find the areas of the bounded regions.

Zadanie 16:

Exercises 57 to 64 concern the graph of the general quartic:

$$f(x) = ax^4 + bx^3 + cx^2 + dx + e, \qquad a \neq 0.$$

57. Using the substitution $x - b/4a$ for x, show that one can reduce to the case

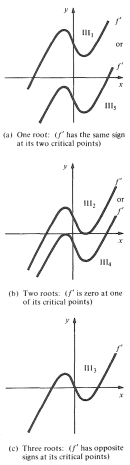
$$f(x) = x^4 + cx^2 + dx + e$$

(with a new c, d and e!).

58. According to the classification of cubics, $f'(x) = 4x^3 + 2cx + d$ can be classified into three types: I ($c > 0$), II ($c = 0$), and III ($c < 0$), so we may name each quartic by the type of its derivative. Sketch the graph of a typical quartic of type I.

59. Divide type II quartics into three cases: II_1 ($d > 0$), II_2 ($d = 0$), II_3 ($d < 0$). Sketch their graphs with $e = 0$.

60. In case III ($c < 0$), $f'(x)$ has two turning points and can have one, two, or three roots. By considering Fig. 3.4.23, show that the sign of $f'(x)$ at its critical points determines the number of roots. Obtain thereby a classification of type III quartics into five subtypes III_1, III_2, III_3, III_4, and III_5. Sketch the graphs for each case and determine the conditions on c and d which govern the cases.



(a) One root: (f' has the same sign at its two critical points)

(b) Two roots: (f' is zero at one of its critical points)

(c) Three roots: (f' has opposite signs at its critical points)

Figure 3.4.23. The five possible positions of the graph of f' in case III.

61. Using your results from Exercises 59 and 60, show that the (c, d) plane may be divided into regions, as shown in Fig. 3.4.24, which determine the type of quartic.[4]

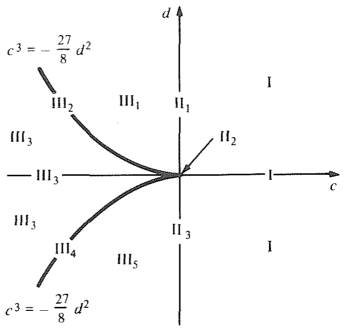


Figure 3.4.24. Locating the value of (c, d) in this graph tells the type of quartic.

Using the results of Exercise 61, classify and sketch the graphs of the quartics in Exercises 62–64.

★62. $x^4 - 3x^2 - 4x$.

★63. $x^4 + 4x^2 + 6x$.

★64. $x^4 + 7x$.

Zadanie 17:

18. A paint can is kicked off the roof of a 156-foot building, its distance S from the ground being given by $S = 156 + 4t - 16t^2$ (S in feet, t in seconds).

(a) How high up does the can go before it falls downward?

(b) At what time does the can return to roof level?

(c) Find the velocity of the can when it collides with the ground.

Zadanie 18:

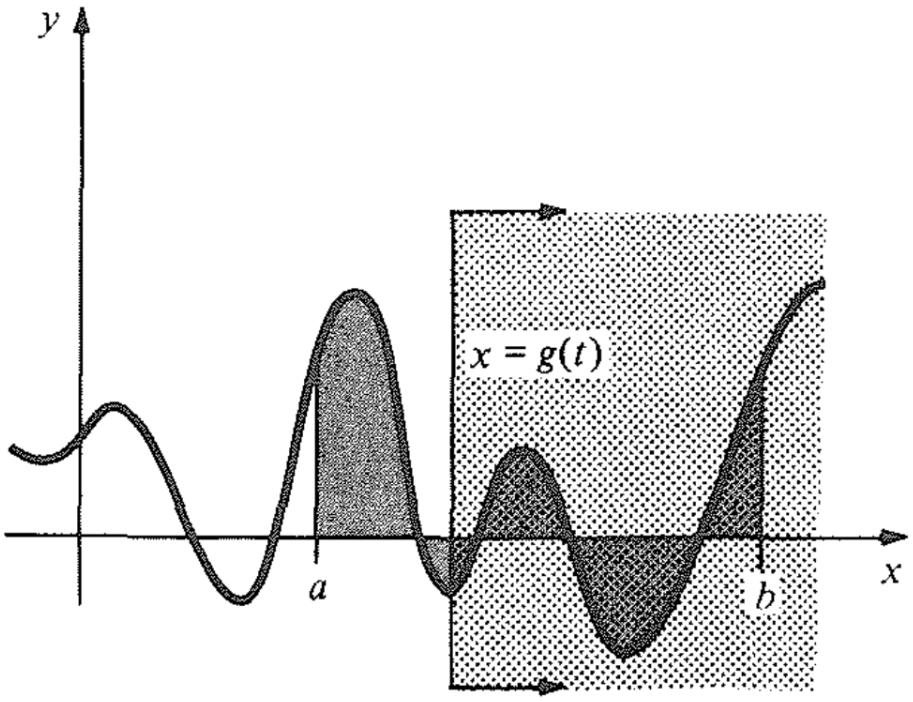
52. Find $\frac{d}{ds}\int_0^{s^2} \frac{1}{\sqrt[3]{x^2+1}}\,dx$. (*Hint*: See Exercise 43, Section 4.5.)

43.

(a) Combine the alternative version of the fundamental theorem of calculus with the chain rule to prove that

$$\frac{d}{dt}\int_a^{g(t)} f(s)\,ds = f(g(t))\cdot g'(t).$$

(b) Interpret (a) in terms of Fig. 4.5.5.



Zadanie 19:

45. If $f(x)$ is positive for all x, do $f(x)$ and $1/f(x)$ have the same inflection points?

Zadanie 20:

108. (a) Prove that if f/g is a rational function (i.e., a quotient of polynomials) with derivative zero, then f/g is a constant.
(b) Conclude that if the rational functions F and G are both antiderivatives for a function h, then F and G differ by a constant.